D3.2 Inheritance

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-d3-2.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 747–759 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Two genes

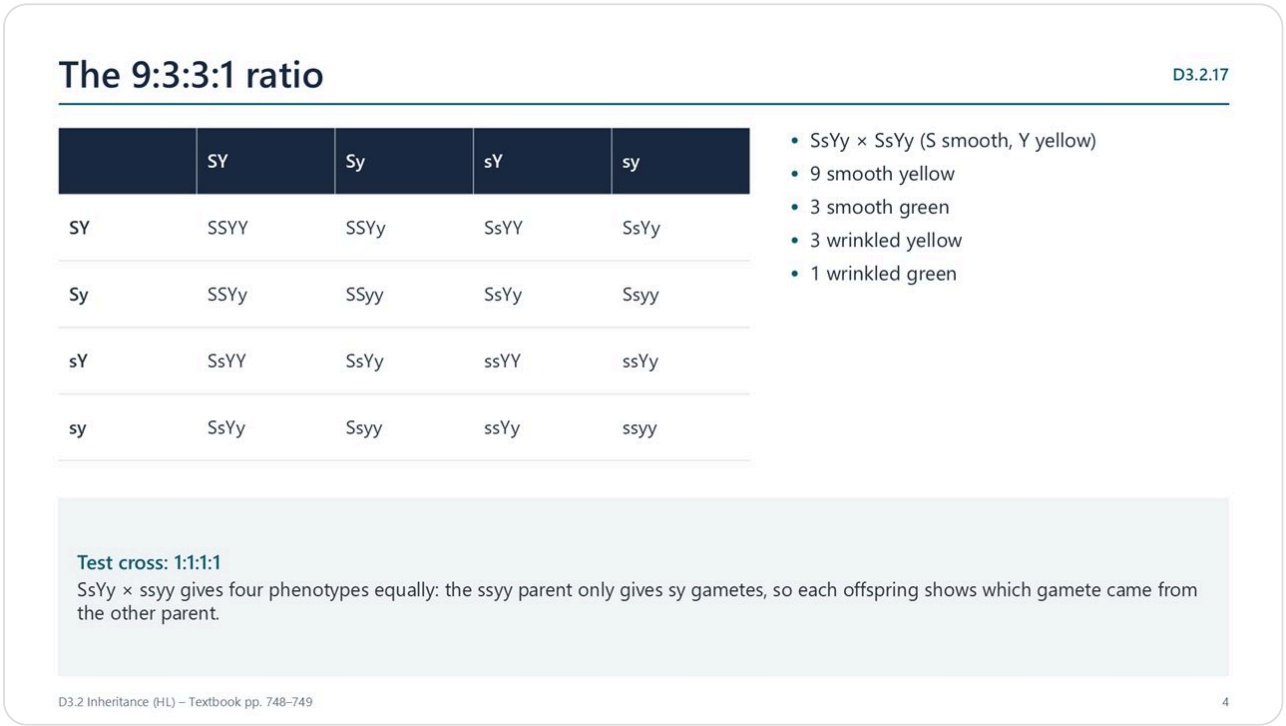
D3.2.16 Segregation and independent assortment

pp. 747–748

In meiosis I, homologous chromosomes separate, so each gamete gets one allele of each gene (**segregation**). Each pair of homologues lines up at random in metaphase I, so genes on different chromosomes are inherited independently (**independent assortment**). A parent RrYy makes RY, Ry, rY and ry gametes in equal numbers.

D3.2.17 Dihybrid Punnett grids

pp. 748–750



Crossing two double heterozygotes for unlinked genes gives a **9:3:3:1** phenotype ratio. A test cross of a double heterozygote with a double homozygous recessive gives **1:1:1:1**.

D3.2.18 Gene loci

pp. 750–751

Databases give each human gene's locus and product. HBB (beta globin) and HBD (delta globin) lie next to each other on chromosome 11; HBA1 (alpha globin) is on chromosome 16; CFTR is on chromosome 7.

Linkage and statistics

D3.2.19 Linkage

pp. 751–752

Genes on the same chromosome are **linked** and tend to be inherited together, so they do not assort independently. They are shown with alleles beside vertical lines, one line per homologous chromosome. The closer the loci, the stronger the linkage.

D3.2.20 Recombinants

pp. 752–754

A **recombinant** has a combination of alleles not found in either parent. With linked genes, recombinants come only from crossing over between the loci, so a test cross gives two large parental classes and two small recombinant classes.

D3.2.21 Chi-squared test

pp. 755–757

The null hypothesis is that observed numbers do not differ significantly from those expected (for example 9:3:3:1). Calculate $\chi^2 = \Sigma(O - E)^2/E$, with degrees of freedom = classes − 1. If χ^2 is greater than the critical value at $p = 0.05$ (7.81 for 3 degrees of freedom), reject the null hypothesis; the genes may be linked.

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